PLC / Machine Data Ingest — Integration Specification

For the JP Minda PLC / automation engineer — pushing live machine data into the tracking system

Version: 1.0   **Date:** 06 Jul 2026   **Audience:** PLC / automation team

This is how a machine (or an edge gateway) sends live cylinder-process data into the CNG tracking system. The endpoint is deliberately **tolerant** — send snake_case or camelCase, one reading or a batch — but this document defines the recommended shape.

1 Endpoint

Production	`POST https://digitisationapi.jpmgroup.co.in/api/v1/cng/ingest`
Content-Type	`application/json`
Auth header	`x-api-key: <the key issued to you>`
Response	`202 Accepted` for any valid JSON — **the line never stalls**. A bad/missing key returns `401`.

The API key is issued per line/gateway and is revocable. It is **not** the SAP sync key — it is a dedicated CNG ingest key.

2 Message shape (one machine reading per message)

```
{
  "machineId":  "HT-401",                 // REQUIRED — which machine reported
  "machineName":"Heat Treatment 1",       // optional — friendly name
  "stageNo":    6,                        // 1–21, the process stage this machine feeds
  "pipeId":     "JPM24C08921",            // the cylinder being processed (see §4)
  "timestamp":  "2026-07-04T09:41:22Z",   // ISO-8601; if omitted, arrival time is used
  "data": {                               // the actual readings + verdict
    "furnace_temp": 892,
    "soak_min":     42,
    "result":       "OK",                 // "OK" | "NG" → pass/fail for this stage
    "status":       "run"                 // run | idle | stopped | fault
  }
}
```

Batches: you may also POST a **JSON array** of such objects in one request.

3 Field guide (all aliases accepted — first match wins)

Field	Required	Accepted keys	Notes
Machine ID	✓	`machineId`, `machine_id`, `deviceId`, `tag`, `machine`	uniquely identifies the machine/station
Pipe ID	✓ (stages 2–21)	`pipeId`, `pipe_id`, `barcode`, `serial`, `srNo`, `cylinderId`	the cylinder this reading belongs to (see §4)
Stage No	recommended	`stageNo`, `stage_no`, `stage`	1–21; if omitted, the machine's pre-mapped stage is used
Timestamp	optional	`timestamp`, `ts`, `time`, `deviceTs`	ISO-8601 preferred
Readings	✓	`data`, `readings`, `params`, `metrics`, `values`	an object of `{ name: value }`; OR put the readings as bare top-level keys
Result	recommended	`result` inside `data`	`"OK"` = pass, anything else = fail (raises a defect)
Status	optional	`status` inside `data`	`run` / `idle` / `stopped` / `fault` — drives the machine board

Fault sentinel values `-32768, 32767, 65535, -1` are auto-detected and shown as **FAULT**, never as data.

4 How the cylinder is identified (important)

Every cylinder has a unique **Pipe ID**, created at **Stage 1 (Pipe Cutting)** by the operator. For a reading to attach to the right cylinder, the machine must tell us which Pipe ID it's working on, in **one** of two ways:

1. **Inline (preferred):** include `pipeId` in every message (as above). Simplest if the PLC/gateway knows the cylinder on the machine.
2. **Scan-in:** if the PLC can't send an ID, the operator scans the cylinder onto the machine first (a one-time `POST /api/v1/cng/scan`), and subsequent readings auto-attach to that machine's active cylinder.

Without a Pipe ID a reading is still accepted and stored, but it will show as machine-status only (not attached to a cylinder's trace).

5 Examples

A normal stage reading (Heat Treatment, pass):

```
{ "machineId":"HT-401", "stageNo":6, "pipeId":"JPM24C08921",
  "data": { "furnace_temp":892, "soak_min":42, "result":"OK", "status":"run" } }
```

A rejected reading (Air Leak Test, fail → raises a defect):

```
{ "machineId":"ALT-1301", "stageNo":13, "pipeId":"JPM24C08921",
  "data": { "leak_rate":0.34, "result":"NG", "status":"fault" } }
```

curl test:

```
curl -X POST https://digitisationapi.jpmgroup.co.in/api/v1/cng/ingest \
  -H "Content-Type: application/json" \
  -H "x-api-key: <YOUR_KEY>" \
  -d '{"machineId":"HT-401","stageNo":6,"pipeId":"JPM24C08921","data":{"furnace_temp":892,"result":"O
K","status":"run"}}'
```

6 Checklist for the PLC engineer

- ☐ Confirm the **field names** your PLC actually emits (send us one real sample payload).
- ☐ Confirm the **machine → stageNo** mapping for each station (1–21).
- ☐ Confirm how the **Pipe ID** reaches each machine (inline vs scan-in).
- ☐ Confirm the **result** flag values used for pass/fail.
- ☐ We issue you the `x-api-key`; you POST to the endpoint above.